

Chapter 3: Dynamics

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In this chapter, we turn our attention to the causes of motion: What makes an object change its motion? For centuries, philosophers believed that an object must be continuously acted upon in order to remain in motion. This view was consistent with everyday experience. For example, when you slide a book across a table, it eventually comes to rest. It therefore seems natural to conclude that a “push” is required to keep the book moving.

Galileo Galilei challenged this idea by introducing the principle of inertia. He argued that an object does not need a cause to maintain a constant velocity. In the absence of external influences, an object at rest remains at rest, while an object in motion continues to move in a straight line at constant speed. The apparent need for a cause to maintain motion in everyday situations arises because friction and other external influences are present.

Isaac Newton later formulated this idea and developed a quantitative theory of dynamics. In particular, Newton’s asserted that there must be a cause for objects to accelerate. He called this cause *force*. Thus, the fundamental question of dynamics is not why an object has a velocity, but why its velocity changes.

In this Chapter, we will restrict our attention to the motion of a single particle. We will call this particle a *system* to make our terminology consistent with the rest of the text where we will be revisiting the dynamics for system of particles in Chapter ??.

1 Newton’s Axioms of Motion

Axiom 1: A body remains at rest or continues to move in a straight line with constant speed unless acted upon by a *force*.

Axiom 2: A body acted upon a *force* moves in such a manner that the time rate of change of its *momentum* equals the net force acting on it.

Axiom 3: To every action, there is an equal and opposite reaction.

It looks like the 1st axiom is a special case of the 2nd axiom, where the net force is zero. However, the 1st axiom is important because Newton defines a special class of reference frames

called *inertial frames* where his axioms of motion hold true. Geometrically, a *reference frame* is the coordinate system where we make measurements. For example, if you are in a car, the car itself is your reference frame; you measure all lengths relative to the car. When the car accelerates, you feel a force pushing you back into your seat. This force is not due to any physical interaction with the car; rather, it arises because you are in a non-inertial frame of reference. Newton's axioms are valid only in inertial frames, which are reference frames that are not accelerating. Inertial frames are special because they allow us to describe the motion of objects in a simple and consistent way.

The second axiom mentions the quantity *momentum*, which is defined as

$$\mathbf{p} = m\mathbf{v}, \quad (1)$$

where $\mathbf{p}$ is the momentum of a body, m is its mass, and $\mathbf{v}$ is its velocity. The units of momentum are kilogram meters per second (kg·m/s). The second axiom states that if the momentum changes in time it is due to a net force acting on the body. Mathematically, this can be expressed as

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}, \quad (2)$$

where $\mathbf{F}$ is the net force acting on the body. The units of force are newtons (N), which are defined as one kilogram meter per second squared (kg·m/s²). In general, both mass and velocity can change with time resulting in a change in momentum. If the mass of the body is constant, then this equation can be rewritten as

$$\mathbf{F} = m \frac{d\mathbf{v}}{dt} = m\mathbf{a}, \quad (3)$$

where $\mathbf{a}$ is the acceleration of the body. This equation is often referred to as Newton's second law of motion. It states that the acceleration of a body is directly proportional to the net force acting on it and inversely proportional to its mass. In other words,

$$\mathbf{a} = \frac{\mathbf{F}}{m}. \quad (4)$$

The third axiom states the nature of forces that they never exist in isolation, but always come in pairs. If body A exerts a force on body B, then body B exerts an equal and opposite force on body A. This is known as the *action-reaction* principle. For example, when you push against a wall, the wall pushes back against you with an equal and opposite force. If you push a book across a table, the book pushes back against your hand with an equal and opposite force.

These three axioms explain a wide range of phenomena, from the motion of planets to the behavior of everyday objects. They form the foundation of classical mechanics and are still used today to analyze the motion of objects in a variety of contexts.

2 Galilean Invariance and Inertial Frames

All reference frames that move with a constant velocity relative to an inertial reference frame are also inertial reference frames. The laws of mechanics have the same form in all inertial reference frames. This is known as *Galilean invariance*. For example, if you are in a car moving at constant velocity along a straight road, you can perform mechanical experiments and obtain the same results as if you were at rest. The outcome of a mechanical experiment does not depend on the uniform motion of the reference frame.

To see this, consider two inertial reference frames, S and S' , where S' moves with a constant velocity $\mathbf{v}_0$ relative to S . If their origins coincide at $t = 0$, the position of a body in the two reference frames is related by

$$\mathbf{r}' = \mathbf{r} - \mathbf{v}_0 t, \quad (5)$$

where $\mathbf{r}$ is the position of the body in S and $\mathbf{r}'$ is its position in S' . Differentiating with respect to time gives the relation between the velocities:

$$\mathbf{v}' = \frac{d\mathbf{r}'}{dt} = \frac{d\mathbf{r}}{dt} - \mathbf{v}_0 = \mathbf{v} - \mathbf{v}_0, \quad (6)$$

where $\mathbf{v}$ and $\mathbf{v}'$ are the velocities of the body in S and S' , respectively. Differentiating once more gives

$$\mathbf{a}' = \frac{d\mathbf{v}'}{dt} = \frac{d\mathbf{v}}{dt} = \mathbf{a}. \quad (7)$$

Thus, the acceleration of a body is the same in all inertial reference frames. Consequently, Newton's second law,

$$\mathbf{F} = m\mathbf{a}, \quad (8)$$

has the same form in both frames:

$$\mathbf{F}' = m\mathbf{a}' = m\mathbf{a} = \mathbf{F}. \quad (9)$$

This invariance of Newton's laws under Galilean transformations is the basis of Galilean invariance in classical mechanics.

An important feature of Galilean transformations is that time is absolute. The time interval between two events is the same in all inertial reference frames:

$$t' = t. \quad (10)$$

Therefore, if two events occur simultaneously in one inertial reference frame, they occur simultaneously in every other inertial reference frame. This absolute nature of time is a fundamental assumption of classical mechanics.

3 Validity of Newton's Axioms

The validity of Newton's axioms has been extensively tested and confirmed by experiments. Within the domain of classical mechanics, they provide an accurate description of the motion of macroscopic bodies when their speeds are much smaller than the speed of light and quantum effects can be neglected.

The first and second axioms form the basis for relating forces to the motion of a particle, while the third axiom expresses a fundamental property of interactions between two bodies. In its usual form,

$$\mathbf{F}_{12} = -\mathbf{F}_{21},$$

the third axiom holds for central interactions, for which the force between two particles acts along the line joining them and depends only on their separation. Newtonian gravitational and electrostatic forces are important examples. By contrast, forces such as magnetic forces, friction, and fluid drag are non-central: their direction is not determined solely by the line joining the interacting bodies. Nevertheless, non-central forces do not necessarily violate the third axiom; in interactions involving fields, the momentum of the field must also be taken into account.

The axioms therefore constitute the postulates of Newtonian mechanics: their validity is established not by derivation from more fundamental principles within classical mechanics, but by their agreement with experimental observations within their range of applicability.

4 Measurement of Mass

An *isolated system* is a system on which the net external force is zero. Internal forces may still act between the bodies within the system. Therefore, for an isolated system consisting of N bodies, the sum of the internal forces acting on all bodies is zero:

$$\sum_{i=1}^N \mathbf{F}_i = 0. \tag{11}$$

For a system consisting of two bodies, Newton's third axiom gives

$$\mathbf{F}_1 = -\mathbf{F}_2. \quad (12)$$

Using Newton's second axiom,

$$m_1 \mathbf{a}_1 = -m_2 \mathbf{a}_2, \quad (13)$$

and therefore, taking magnitudes,

$$\frac{m_2}{m_1} = \frac{|\mathbf{a}_1|}{|\mathbf{a}_2|}. \quad (14)$$

Thus, if one of the masses is chosen as a reference mass, the other mass can be determined by measuring the accelerations of the two bodies.

There is another way to measure mass. Near the surface of the Earth, an object is acted upon by the gravitational force, called its *weight*. Neglecting air resistance, all objects at the same location have the same gravitational acceleration, denoted by $\mathbf{g}$. Newton's second axiom therefore gives

$$\mathbf{W} = m\mathbf{g}, \quad (15)$$

where $\mathbf{W}$ is the weight of the object. Taking magnitudes,

$$|\mathbf{W}| = m|\mathbf{g}|. \quad (16)$$

Thus, by comparing the weights of two bodies at the same location, their masses are related by

$$\frac{m_2}{m_1} = \frac{|\mathbf{W}_2|}{|\mathbf{W}_1|}, \quad (17)$$

or equivalently,

$$m_2 = \frac{|\mathbf{W}_2|}{|\mathbf{W}_1|} m_1. \quad (18)$$

The equality of inertial and gravitational mass is an experimental fact in Newtonian mechanics and is known as the *equivalence of inertial and gravitational mass*. This equivalence plays a central role in Einstein's theory of gravity, where it is elevated to the *equivalence principle*.

5 Equations of Motion

The 2nd axiom provides the *equation of motion* of a body once the force acting on it is specified. It is a differential equation that determines how the position of the body changes with time under the influence of a force. In general,

$$\dot{\mathbf{p}} = \mathbf{F}(\mathbf{r}, \mathbf{v}, t), \quad (19)$$

where $\mathbf{r}$ is the position vector, $\mathbf{v}$ is the velocity vector, and t is time. The force may depend on position, velocity, and time.

For a body of constant mass, $\mathbf{p} = m\mathbf{v}$, so the equation of motion takes the familiar form

$$m\ddot{\mathbf{r}} = \mathbf{F}(\mathbf{r}, \dot{\mathbf{r}}, t). \quad (20)$$

This is a second-order differential equation for the position vector $\mathbf{r}(t)$. Together with the initial position and velocity, its solution determines the motion of the body.

5.1 Angular-Momentum Form of the 2nd Axiom

The 2nd axiom also has a useful form associated with rotational motion. In translational motion, the momentum $\mathbf{p}$ characterizes the motion of a body, while the force $\mathbf{F}$ determines how its momentum changes. There is a corresponding description of motion about a reference point: the angular momentum $\mathbf{L}$ characterizes the motion of a body about that point, while the torque $\boldsymbol{\tau}$ determines how its angular momentum changes. Thus, angular momentum and torque play roles in rotational motion analogous to those of momentum and force in translational motion.

The angular momentum of a particle about a reference point O is defined as

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}, \quad (21)$$

where $\mathbf{r}$ is the position vector of the particle relative to O . The units of angular momentum are kilogram meter squared per second ($\text{kg}\cdot\text{m}^2/\text{s}$). The torque of a force about the same point is defined as

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}. \quad (22)$$

These definitions reflect the fact that a force can affect not only the translational motion of a body but also its tendency to rotate about a reference point. The torque depends on both the magnitude and direction of the force and on its position relative to the reference point.

Taking the time derivative of the angular momentum and using the 2nd axiom gives

$$\frac{d\mathbf{L}}{dt} = \frac{d\mathbf{r}}{dt} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt}. \quad (23)$$

For a particle of constant mass, $\mathbf{p} = m\mathbf{v}$, so $\mathbf{v}$ and $\mathbf{p}$ are parallel and therefore

$$\mathbf{v} \times \mathbf{p} = 0. \quad (24)$$

Hence,

$$\boxed{\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}}. \quad (25)$$

Thus, the 2nd axiom can be written in two analogous forms,

$$\boxed{\frac{d\mathbf{p}}{dt} = \mathbf{F}}, \quad \boxed{\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}}. \quad (26)$$

The first describes the change of linear momentum caused by a force, while the second describes the change of angular momentum caused by a torque. These relations will be used in the following section to formulate the conservation laws.

6 Conservation Theorems

In this section, we will discuss the direct consequences of Newton's axioms of motion, which are the conservation theorems. These theorems state that certain physical quantities remain constant in time for an isolated system; this is what we mean by *conservation*. The conservation theorems are important because they provide a powerful tool for analyzing the motion of systems without having to solve the equations of motion directly. The conservation theorems are derived from Newton's axioms and are valid for all isolated systems, regardless of the specific forces acting within the system.

6.1 Conservation of Linear Momentum

The linear momentum of a system is conserved if the net external force acting on the system is zero:

$$\dot{\mathbf{p}} = \mathbf{F}_{\text{ext}} = 0 \implies \mathbf{p} = \text{constant}, \quad (27)$$

The system here can be a single particle or a collection of particles. The conservation of linear momentum says that it does not matter how a single particle moves in space, or how

the particles move in space and interact with each other, as long as there is no net external force acting on the system the total linear momentum of the system remains constant.

A corollary of the conservation of linear momentum is that if it is conserved in a system, so all components of linear momentum vector are conserved. This means that if the system is isolated, the total linear momentum in the x -direction, y -direction, and z -direction are all conserved separately. To see this, let s be any arbitrary direction in space, and let $\hat{\mathbf{s}}$ be the unit vector in that direction. The component of the linear momentum along this direction is given by

$$p_s = \mathbf{p} \cdot \hat{\mathbf{s}}. \quad (28)$$

Taking the time derivative gives

$$\dot{p}_s = \dot{\mathbf{p}} \cdot \hat{\mathbf{s}} = \mathbf{F}_{\text{ext}} \cdot \hat{\mathbf{s}} = F_{\text{ext},s}, \quad (29)$$

where $F_{\text{ext},s}$ is the component of the net external force along the direction $\hat{\mathbf{s}}$. If the net external force is zero, then $F_{\text{ext},s} = 0$, and therefore $\dot{p}_s = 0$. This means that the component of the linear momentum along any arbitrary direction is conserved. Since the x -direction, y -direction, and z -direction are just three specific directions in space, it follows that the components of linear momentum along these directions are also conserved separately.

6.2 Conservation of Angular Momentum

The angular momentum of a system is conserved if the net external torque acting on the system is zero.

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}_{\text{ext}} = 0 \implies \mathbf{L} = \text{constant}, \quad (30)$$

6.3 Work and Kinetic Energy

When a particle is acted upon by a force, $\mathbf{F}$, the *work* done by the force as the particle moves from position $\mathbf{r}_1$ to position $\mathbf{r}_2$ is defined as

$$W = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r}. \quad (31)$$

where $d\mathbf{r}$ is an infinitesimal displacement vector along the path of the particle from $\mathbf{r}_1$ to $\mathbf{r}_2$ and it can be expressed as

$$d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k} = \mathbf{v}dt, \quad (32)$$

If $\mathbf{F}$ is the only force acting on the particle, then we can write the integrand of the work integral as

$$\mathbf{F} \cdot d\mathbf{r} = \frac{d\mathbf{p}}{dt} \cdot d\mathbf{r} = \frac{d\mathbf{p}}{dt} \cdot \mathbf{v}dt = \frac{d\mathbf{p}}{dt} \cdot \frac{\mathbf{p}}{m}dt = \frac{1}{m}\mathbf{p} \cdot d\mathbf{p} = \frac{1}{2m}d(\mathbf{p} \cdot \mathbf{p}) = d\left(\frac{1}{2}mv^2\right) = dK, \quad (33)$$

where $K = \frac{1}{2}mv^2$ is defined as the *kinetic energy* of the particle. Therefore, the work done by the force as the particle moves from $\mathbf{r}_1$ to $\mathbf{r}_2$ can also be expressed as the change in kinetic energy of the particle:

$$W_{12} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \int_{K_1}^{K_2} dK = K_2 - K_1 = \Delta K. \quad (34)$$

From the differential point of view, if a force $\mathbf{F}$ acts on a particle along an infinitesimal path $d\mathbf{r}$, it changes its kinetic energy by an infinitesimal amount dK :

$$dK = \mathbf{F} \cdot d\mathbf{r}. \quad (35)$$

Both equations, Eq. (34) and Eq. (35), are known as the *work-energy theorem*. They state that the work done by a force on a particle is equal to the change in its kinetic energy. This theorem provides a powerful tool for analyzing the motion of particles, especially when the forces involved are complex or difficult to describe in terms of acceleration.

It is worth emphasizing that a positive work done on a particle by a force increases its kinetic energy, while a negative work done by a force decreases its kinetic energy. If the work done by the force is zero, then the kinetic energy of the particle remains constant. This can happen if the force is perpendicular to the displacement of the particle, as in uniform circular motion, or if there is no net force acting on the particle.

The unit of work and kinetic energy is the joule (J), which is defined as one newton-meter (N·m). In terms of base SI units, one joule is equal to one kilogram meter squared per second squared ($\text{kg}\cdot\text{m}^2/\text{s}^2$).

6.4 Conservative Forces and Potential Energy

A *conservative force* is a force for which the work done in moving a particle between two points is independent of the path taken. Thus, the work depends only on the initial and final positions of the particle. Equivalently, the work done by a conservative force around any closed path is zero:

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0. \quad (36)$$

This property allows us to define a scalar function of position called the *potential energy*, $U(\mathbf{r})$, such that the work done by the conservative force in moving the particle from point A to point B is equal to the negative change in potential energy:

$$W_{AB} = \int_{\mathbf{r}_A}^{\mathbf{r}_B} \mathbf{F} \cdot d\mathbf{r} = -\Delta U = U(\mathbf{r}_A) - U(\mathbf{r}_B). \quad (37)$$

For simplicity, consider motion in the x direction only. Then,

$$W_{AB} = \int_{x_A}^{x_B} F_x dx = U(x_A) - U(x_B). \quad (38)$$

Since this relation must hold for arbitrary x_A and x_B , differentiating with respect to the upper limit gives

$$F_x = -\frac{dU}{dx}. \quad (39)$$

In three dimensions, this generalizes to

$$\mathbf{F} = -\nabla U, \quad (40)$$

where ∇U is the gradient of the potential energy:

$$\nabla U = \frac{\partial U}{\partial x} \mathbf{i} + \frac{\partial U}{\partial y} \mathbf{j} + \frac{\partial U}{\partial z} \mathbf{k}. \quad (41)$$

Thus, the force is determined by how the potential energy varies in space, while the work between two points depends only on the values of U at those points. The zero of potential energy is arbitrary: adding a constant to U does not affect the force or the work:

$$\mathbf{F} = -\nabla(U + C) = -\nabla U \quad (42)$$

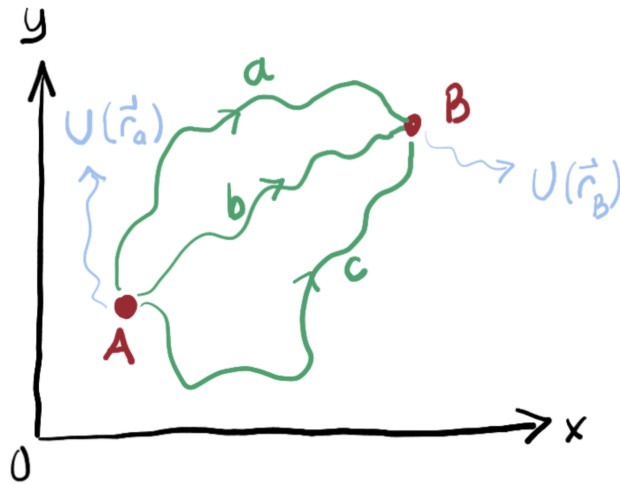


Figure 1: For a conservative force, the work done in moving a particle between two points is independent of the path taken; it depends only on the initial and final positions. Therefore, the work done around any closed path is zero.

6.5 Conservation of Mechanical Energy

The *mechanical energy* of a particle is defined as the sum of its kinetic energy and potential energy:

$$E = K + U, \quad (43)$$

where K is the kinetic energy and U is the potential energy. A very important consequence of Newton's axioms is that the mechanical energy of a particle is conserved if the only forces acting on it are conservative forces. In other words, if a particle is acted upon only by conservative forces, then its total mechanical energy remains constant. We will show this mathematically as follows:

$$\frac{dE}{dt} = \frac{dK}{dt} + \frac{dU}{dt} \quad (44)$$

The first term in the summation can be written as

$$\frac{dK}{dt} = \frac{d}{dt} \left(\frac{1}{2} m \mathbf{v} \cdot \mathbf{v} \right) = m \mathbf{v} \cdot \dot{\mathbf{v}} = m \mathbf{v} \cdot \mathbf{a} = \mathbf{F} \cdot \mathbf{v} = \mathbf{F} \cdot \dot{\mathbf{r}} \quad (45)$$

For the second term, we will consider the most general case where the potential energy is a function of position and time, $U(\mathbf{r}, t)$. Then, using the chain rule, we can write

$$\frac{dU}{dt} = \sum_{i=1}^3 \frac{\partial U}{\partial x_i} \frac{dx_i}{dt} + \frac{\partial U}{\partial t} = \sum_{i=1}^3 \frac{\partial U}{\partial x_i} \dot{x}_i + \frac{\partial U}{\partial t} = \nabla U \cdot \dot{\mathbf{r}} + \frac{\partial U}{\partial t}. \quad (46)$$

Therefore, we can write the time derivative of the mechanical energy as

$$\frac{dE}{dt} = \mathbf{F} \cdot \dot{\mathbf{r}} + \nabla U \cdot \dot{\mathbf{r}} + \frac{\partial U}{\partial t} = (\mathbf{F} + \nabla U) \cdot \dot{\mathbf{r}} + \frac{\partial U}{\partial t}. \quad (47)$$

If the force is conservative, then $\mathbf{F} = -\nabla U$, so the first term in the summation vanishes, and we are left with

$$\frac{dE}{dt} = \frac{\partial U}{\partial t}. \quad (48)$$

If the potential energy is independent of time, then $\frac{\partial U}{\partial t} = 0$, and we have

$$\frac{dE}{dt} = 0 \implies E = \text{constant}. \quad (49)$$

Thus, the mechanical energy of a particle is conserved if the only forces acting on it are conservative forces and the potential energy is independent of time. This is known as the *conservation of mechanical energy*. It is a powerful tool for analyzing the motion of particles, especially in situations where the forces involved are complex or difficult to describe in terms of acceleration.

7 More on the Total Energy and the Potential Energy

What if we only know the total mechanical energy and the potential energy function of a system? How much information can we extract about the motion of the system without solving any equation of motion?

8 Mechanical Energy as a First Integral of Motion

The conservation of mechanical energy can be used to obtain a first-order equation governing the motion of a particle. If the mechanical energy of a particle is conserved, then

$$E = K + U = \frac{1}{2}mv^2 + U(x) = \text{constant}. \quad (50)$$

This equation is called a *first integral of motion* because it is obtained by integrating the second-order equation of motion once with respect to time. In this sense, energy conservation provides a first-order form of the equation of motion. This equation can be rearranged to give the speed of the particle as a function of position:

$$v(x) = \sqrt{\frac{2}{m} [E - U(x)]}. \quad (51)$$

For one-dimensional motion, we have

$$v = \frac{dx}{dt}, \quad (52)$$

so that

$$dt = \frac{dx}{\pm \sqrt{\frac{2}{m} [E - U(x)]}}. \quad (53)$$

Integrating from (x_0, t_0) to (x, t) gives

$$t - t_0 = \int_{x_0}^x \frac{dx}{\pm \sqrt{\frac{2}{m} [E - U(x)]}}. \quad (54)$$

Thus, if the potential energy $U(x)$ is known, this equation can, in principle, be used to determine the motion $x(t)$. The integral may not have an analytical solution for an arbitrary potential, but it can be evaluated numerically.

For motion in more than one dimension, conservation of energy determines the speed of the particle but not the direction of its velocity. Therefore, the energy equation alone is not sufficient to determine the full trajectory $\mathbf{r}(t)$.

9 Worked Examples

9.1 Circular motion

Basics and why it takes place on a plane. torque and centripetal force. Derivation of the centripetal acceleration formula. Examples of circular motion in everyday life. angular momentum and torque in circular motion. Conservation of angular momentum in circular motion. Examples of conservation of angular momentum in everyday life.

- 9.2 Motion of a Particle under a Constant Force
- 9.3 Motion of a Particle on an Inclined Plane
- 9.4 Free Fall and Terminal Speed
- 9.5 Collisions and Impulse
- 9.6 Nabla as an operator converting a scalar field into a vector field
- 9.7 mgh example